WAKEUP REDESIGN · V0.3

# Calm until 6:00. Impossible to ignore at 6:00.

A quieter, more typographic app for everything except the alarm itself, which becomes the loudest thing on the phone, visually and audibly. Updated with your answers: rings are continuous and rising, you can mute while you solve, and scanning a QR code ends the morning.

## Five biggest weaknesses of v0.2

1 **Cards everywhere**

Three plan cards, a banner and a status pill, all in rounded bordered boxes. Nothing led the eye. **Now:** a flat surface with hairline dividers. Cards are gone from lists.

2 **The hero wasn't one**

The next ring (84px) competed with three 30px plan times, and the same routine appeared twice. **Now:** one 120px time, one line of description, and the routines in a quiet list below.

3 **Flat, generic type**

One bold grotesque at every size, plus 11 to 13px grey labels. **Now:** very light, very large Sora numerals, Inter for text, a seven-step scale and one weight per role.

4 **Amber on everything, and a ringing screen that wasn't special**

Buttons, chips, toggles, icons and banners all used amber, so it stopped meaning "act". The ringing screen reused the same shapes and merged "stop this ring" with "I'm awake". **Now:** amber only for the primary action, selection and the ring. Ringing is full-bleed amber with two clearly scoped actions.

5 **Web-like controls and missing states**

24px day chips (under the 48dp touch target), three different radii, no safe areas, no denied, loading or error states, no motion. **Now:** one radius scale, 48dp targets, system bars, bottom sheets, defined states and a motion system.

## The four things the app must never blur

**1 · Stop this ring**
Solve the math. Sound stops. Next ring still comes.

**2 · Confirm awake**
Scan the QR in the bathroom.

**3 · Cancel the rest**
Happens only because of step 2.

**4 · Arm tomorrow**
Automatic, after step 2 or at the hard stop.

Amber rules mark the two steps that change what happens to the rest of the morning.

## How a ring behaves

| | |
|---|---|
| **Rising sound** | Starts at 30% and climbs to 100% over 60 seconds. Keeps ringing until the ring is solved or the next ring takes over. Volume never resets lower during a morning. |
| **Quiet while solving** | Opening the mission mutes the sound for 60 seconds. Each correct answer restarts the 60 seconds. Run out and the sound returns where it left off. A thin countdown shows the time left. |
| **Stop this ring** | Three math problems. Solving them stops only the current ring. A confirmation screen says when the next one is due. |
| **Confirm awake** | Scan the QR code you stuck in the bathroom. In the example it cancels the 34 remaining rings. Fallback: type a long phrase. |
| **Hard stop and emergency exit** | Rings end at the stop time and the day is logged as missed. Emergency stop: hold for 10 seconds, then type a phrase. Logged. |
| **Honest status** | If an Android setting is missing, Home and Settings say the alarm may not ring on time. We never claim a guarantee. Not yet tested on a real device. |

## Where the effort went

OPENING THE APP

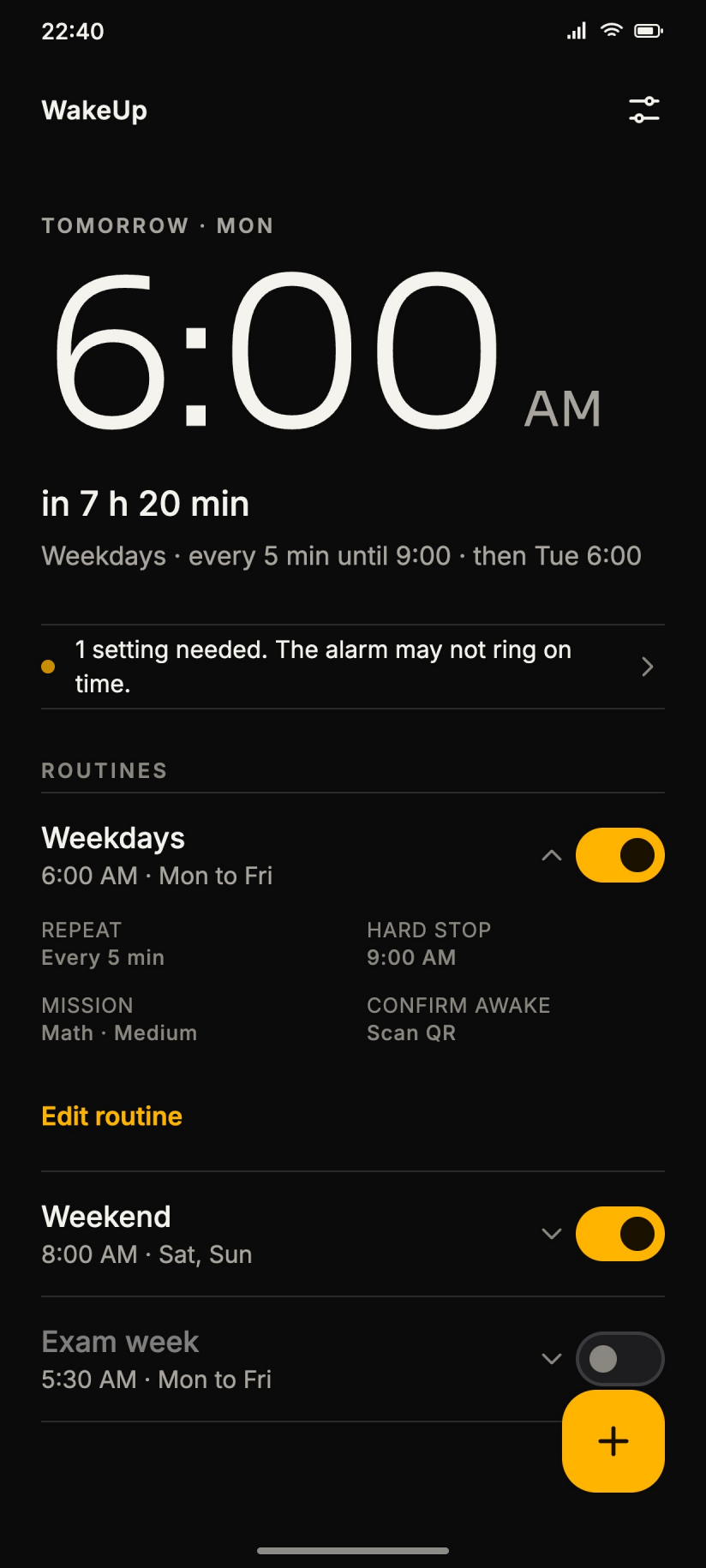
One huge time and one line.

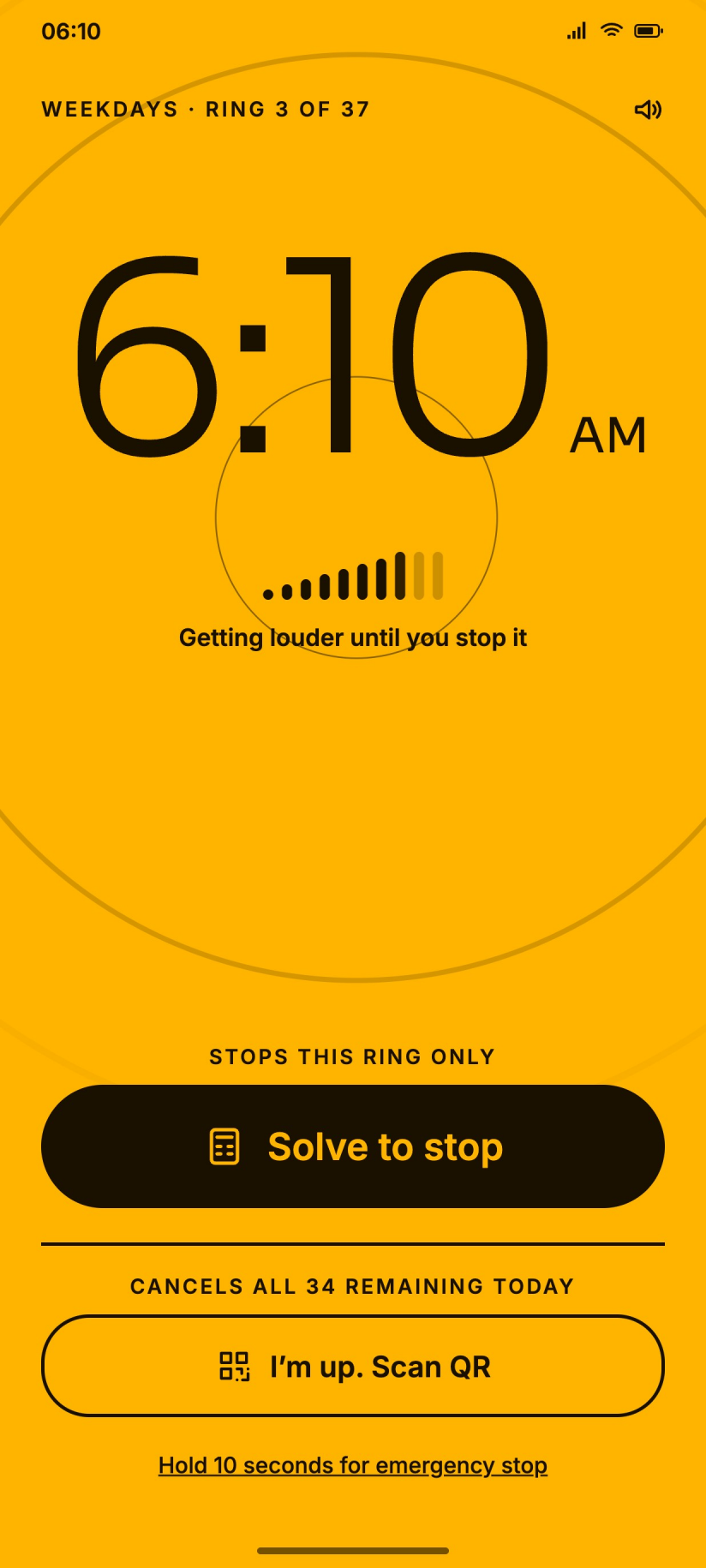
BEING WOKEN UP

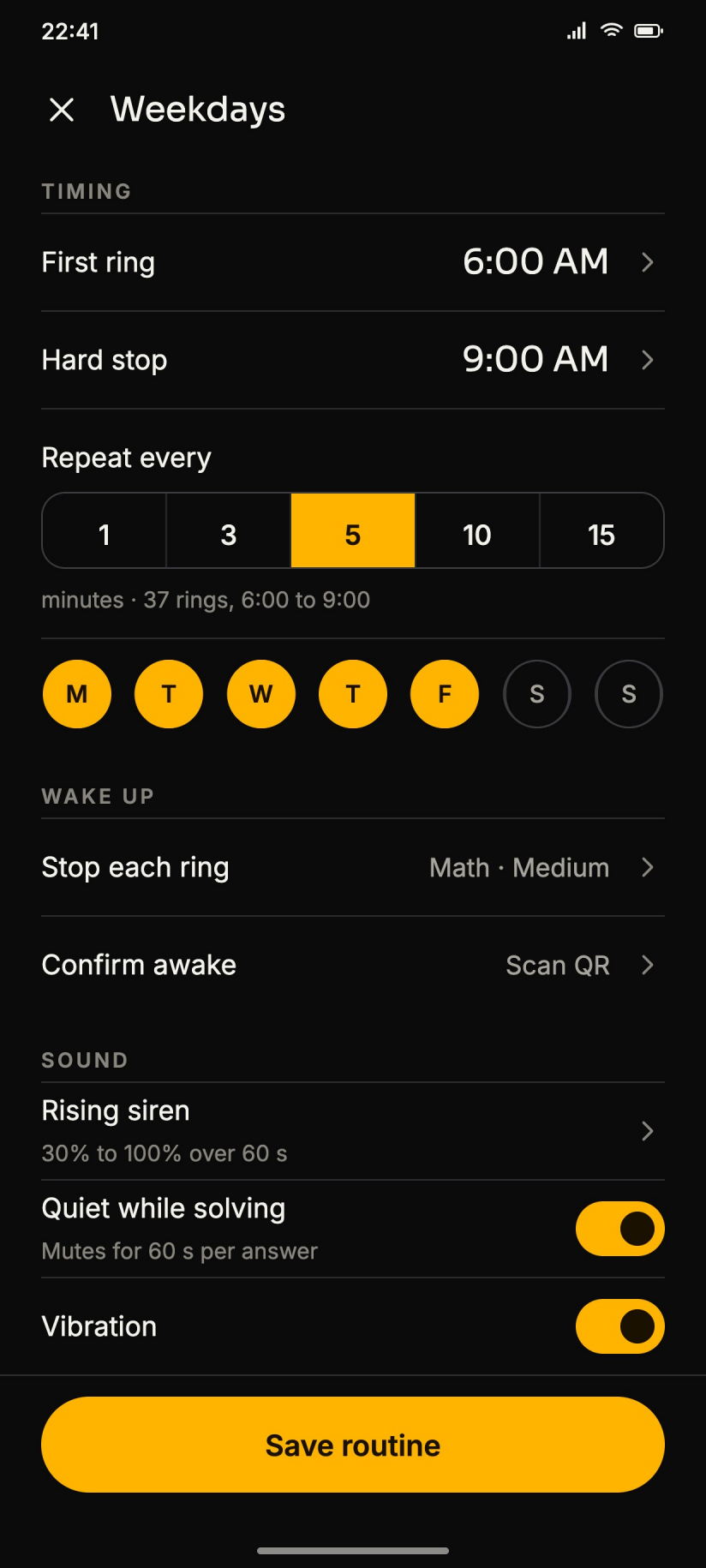
Full-bleed amber, pulsing rings,

FINISHING

A scan that feels like a


22:40
WakeUp
TOMORROW · MON
6:00 AM
in 7 h 20 min
Weekdays · every 5 min until 9:00 · then Tue 6:00
1 setting needed. The alarm may not ring on time.
ROUTINES
Weekdays
6:00 AM · Mon to Fri
REPEAT
Every 5 min
HARD STOP
9:00 AM
MISSION
Math · Medium
CONFIRM AWAKE
Scan QR
Edit routine
Weekend
8:00 AM · Sat, Sun
Exam week
5:30 AM · Mon to Fri


06:10
WEEKDAYS · RING 3 OF 37
6:10 AM
Getting louder until you stop it
STOPS THIS RING ONLY
Solve to stop
CANCELS ALL 34 REMAINING TODAY
I'm up. Scan QR
Hold 10 seconds for emergency stop


22:41
Weekdays
TIMING
First ring
6:00 AM
Hard stop
9:00 AM
Repeat every
1
3
5
10
15
minutes · 37 rings, 6:00 to 9:00
M
T
W
T
F
S
S
WAKE UP
Stop each ring
Math · Medium
Confirm awake
Scan QR
SOUND
Rising siren
30% to 100% over 60 s
Quiet while solving
Mutes for 60 s per answer
Vibration
Save routine

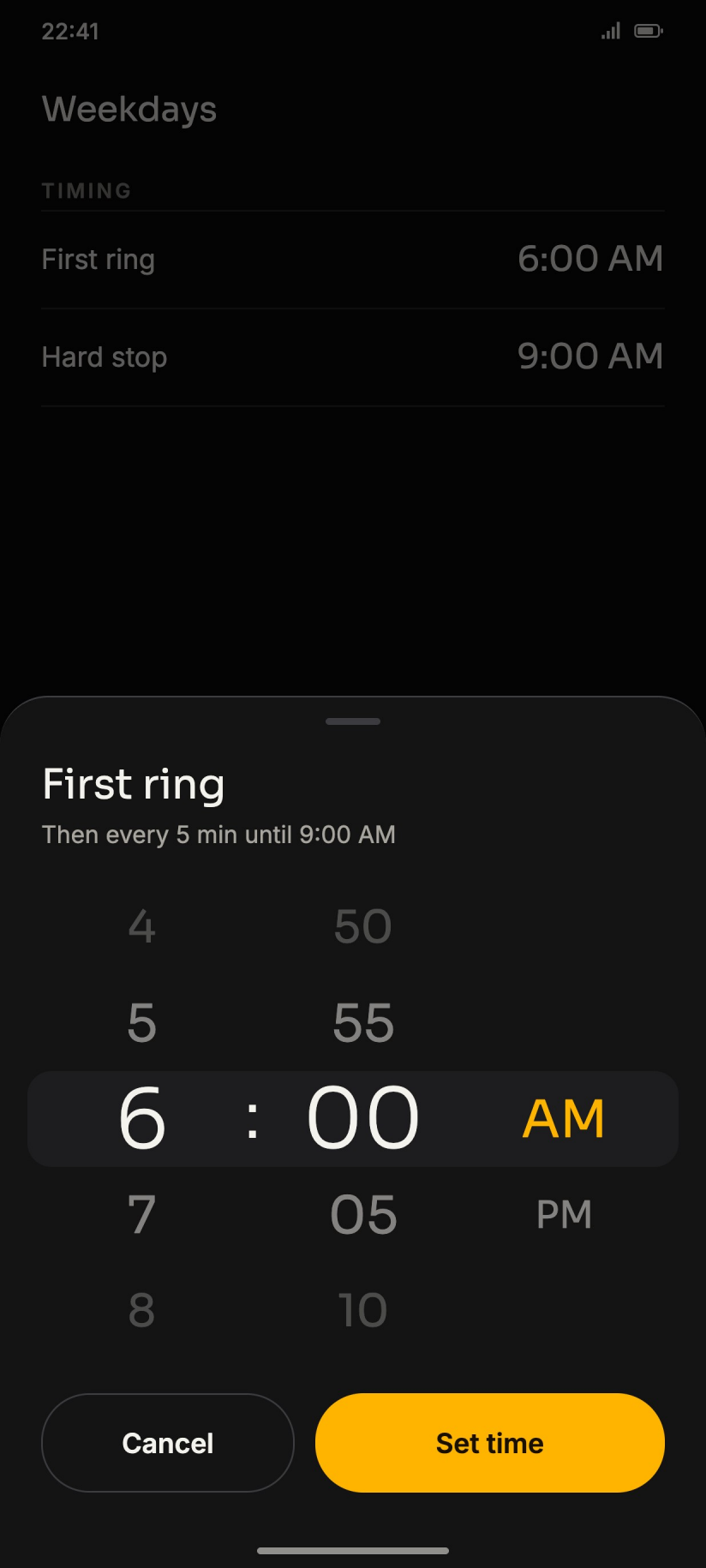
Problem 1 of 3

Sound on

$47 \times 6 - 18$

Sound is back. Keep going

| | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |
| ⌫ | 0 | ✓ |


22:41
Weekdays
TIMING
First ring
6:00 AM
Hard stop
9:00 AM
First ring
Then every 5 min until 9:00 AM
4
50
5
55
6 : 00 AM
7
05
PM
8
10
Cancel
Set time

Ring 3 stopped

NEXT RING

# 6:15 AM

in 4 min

## You’re not marked awake yet

34 more rings are scheduled until 9:00 AM.
Scan your QR code to cancel all of them.

CANCELS ALL REMAINING TODAY

Scan QR to wake up

Close. The next ring will still come

# Missions

Stop a ring | Confirm awake

Solving it silences only the current ring. The next one still comes.

**Math**
Three problems

Easy | Medium | Hard

47 × 6 − 18

**Typing**
Retype a sentence

**Shake**
Shake the phone

LATER

Memory SOON

Steps SOON

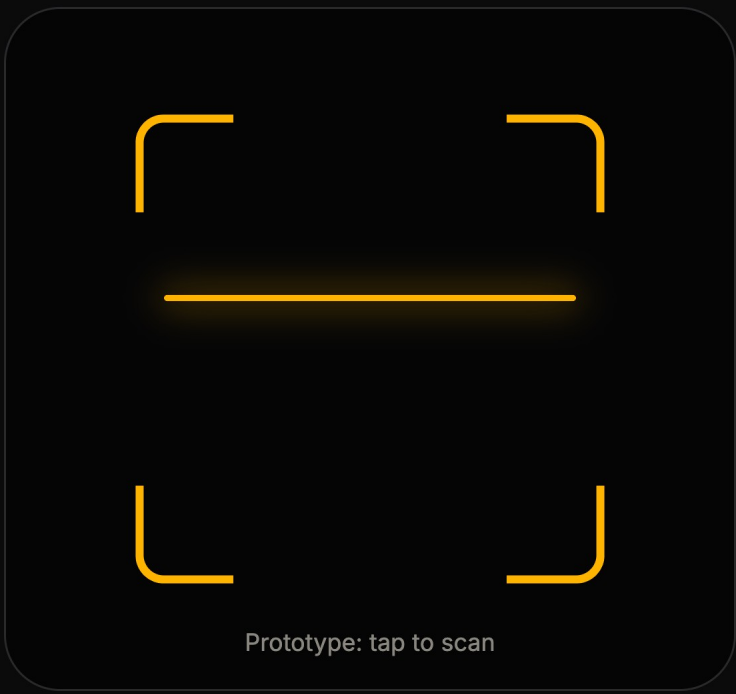
Use Math

FINAL STEP

Ring stopped · now confirm you're awake

# Scan your wake-up code

It's in your bathroom. Walk there, then scan.




**This cancels all 34 remaining rings today**

Tomorrow's 6:00 AM routine arms automatically.

Can't scan? Type the phrase

DESIGN SYSTEM · FOUNDATIONS

# Flat, dark and quiet. Amber means act.

No shadows, no glass, no gradients. Structure comes from hairlines, type size and space. Amber is spent on three things only: the primary action on a screen, a selected state, and the ringing experience.

## Colour

**Background**
#0B0B0C · app background

**Surface 1**
#141415 · sheets, keys

**Surface 2**
#1D1D1F · wheel band, pressed

**Line**
#2A2A2D · hairline dividers

**Line strong**
#3C3C40 · control borders

**Text 1**
#F5F3EE · primary text

**Text 2**
#A8A59E · secondary text

**Text 3**
#8A8780 · hints, captions

**Amber**
#FFB400 · act, selected, ringing

**Amber pressed**
#E09F00 · pressed state

**Amber wash**
#241B06 · tint behind amber

**On amber**
#1A1100 · text on amber

**Success**
#8FD3B0 · correct, allowed

**Error**
#FF8A78 · wrong answer only

Contrast on the background: Text 1 about 17:1, Text 2 about 8:1, Text 3 about 5:1. On amber, On-amber text is about 10:1. Success and error never carry meaning alone: each is paired with an icon or a word.

## Typography

6:10
**Display XL**
Sora 300 · 156/150 · −5%
Ringing time

6:00
**Display L**
Sora 300 · 120/120 · −3%
Home next alarm, ring stopped

47 × 6 − 18
**Display M**
Sora 300 · 60/68 · −2%
Math question

You’re up.
**Headline**
Sora 300 · 32 to 40
Screen statements

Wake plan
**Title**
Sora 400 · 20/28 and 24/30
App bar, sheet titles

Rings land on the minute.
**Body**
Inter 400 · 16/24
Descriptions, rows

**Save routine**
**Label**
Inter 500 to 600 · 14 to 17
Buttons, values, chips

CANCELS ALL REMAINING TODAY
**Caption**
Inter 600 · 12/16 · +14% caps
Section labels, scope tags

Sora carries numerals and headlines (light weights for scale). Inter carries everything you read. Sizes in sp, so the system font scale applies. The two largest specimens are drawn at 50 and 60 percent.

## Spacing

4
8
12
16
24
32
48

4dp base. Screen margin 24. Rows 56 to 72 tall. Every touch target is at least 48 by 48dp. Primary buttons 56, 64 or 72 tall.

## Corner radius

14 · controls, keys
20 · panels, FAB
28 · sheets
Pill · buttons

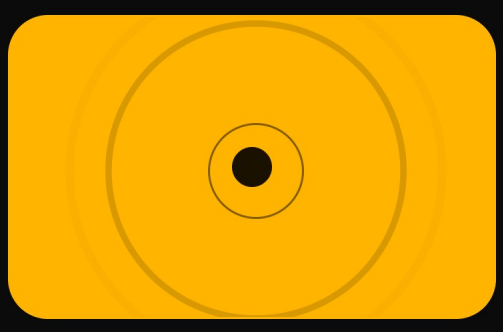
## Icons

24dp grid, 1.75 stroke, round caps and joins, one colour at a time. Never filled, never two-tone.

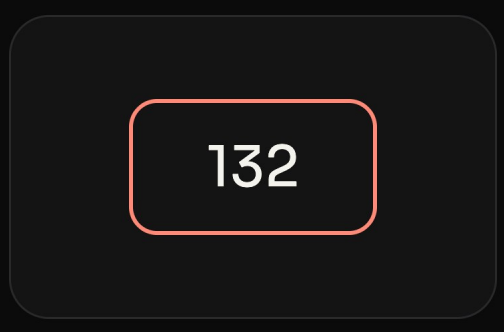
MOTION AND HAPTICS

# Motion explains what changed. Nothing waits for it.

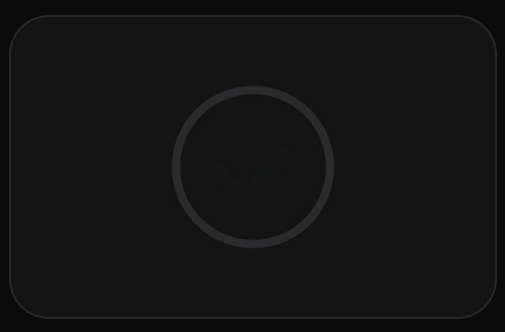
One easing for almost everything: a quick start that settles gently. Four durations. The demos below are live, and they stop when the system asks for reduced motion.



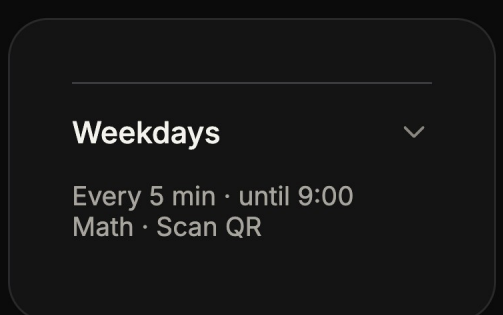
**Pulse rings**
Loops while an alarm rings



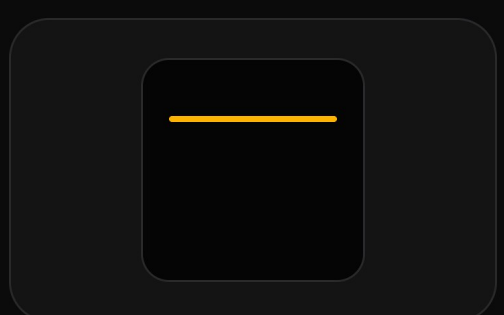

**Wrong answer**
Shake once, border turns coral



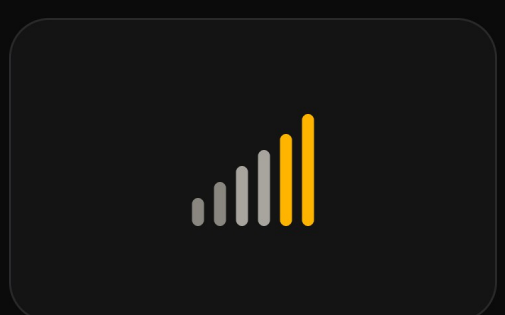
**Day cleared**
Ring draws, then the tick




**Routine details**
Expand and collapse



**QR scan line**
Sweeps while the camera is live



**Volume ramp**
Mirrors the actual gain, 30% to 100%

## Durations

- **90 ms** instant: key press, pressed state
- **160 ms** fast: switch, segment, chip colour
- **240 ms** base: expand, tab, question change, progress
- **400 ms** slow: staggered text on arrival (120 ms steps)
- **700 ms** draw: the completion ring. 360 ms for the tick
- **2400 ms** loop: ring pulse, never faster than 0.4 Hz

## Easing

- **Settle** cubic-bezier(.2, 0, 0, 1) for entering, expanding, pressing
- **Exit** cubic-bezier(.3, 0, 1, 1), 200 ms, sheets only
- **Pulse** cubic-bezier(.2, 0, .2, 1) so each ring eases out
- **Linear** only for the quiet-window countdown
- No bounce, no overshoot. Restraint is the point.

## What animates, when, and how it feels

| ELEMENT | TRIGGER | DURATION | EASING AND MOVEMENT | HAPTIC |
|---|---|---|---|---|
| **Ringing screen** | Alarm fires | 0 ms | Appears instantly, no entrance | Vibration pattern follows the pulse |
| **Pulse rings** | While ringing | 2400 loop | Scale .45 to 2.1, fade .5 to 0, three rings 800 ms apart | None |
| **Volume bars** | Each +10% of gain | 160 | Settle | None |
| **Open mission** | Tap Solve to stop | 240 | Settle. New screen rises 24 px and fades in | Tick |
| **Keypad key** | Touch down | 90 | Scale to .97, Surface 2 | Light tick |
| **Wrong answer** | Check with a wrong value | 360 + 100 | Settle. Shake ±10 px, input clears at 460 ms | Reject, two short buzzes |
| **Correct answer** | Check with a right value | 240 + 520 hold | Settle. 5% pop, segment fills, next question rises 12 px | Confirm |
| **Quiet countdown** | Every second | 1000 | Linear width. At zero the chip turns amber over 240 ms | Single tick when sound returns |
| **Switch, segment** | Tap | 160 | Settle. Thumb slides 20 px and grows | Tick |
| **Routine details** | Tap row | 240 | Settle. Height 0 to auto, chevron turns 180° | Tick |
| **Bottom sheet** | Open and close | 280 / 200 | Settle in, Exit out. Rises 24 px with fade | None |
| **Ring stopped** | Third answer correct | 400 | Settle. Three blocks fade up, 120 ms apart | Confirm |
| **QR scan line** | Camera live | 2000 loop | Ease in-out sweep, fades at the ends | Long success |

WHAT GOES WRONG AT 6 AM

## Edge cases and what the app does

| | |
|---|---|
| **Next ring while one is still ringing** | The old ring stops and the new one starts at the volume already reached. The ring counter goes up. Volume never resets lower in a morning. |
| **Math solved** | Stops this ring only. The next ring is already scheduled, so nothing needs to be re-armed. |
| **Quiet while solving** | The player is set to volume 0, not paused, so the sound comes back instantly. A 60 second timer lives in the service. Each correct answer restarts it. If the app dies, the already-scheduled next ring still fires. |
| **QR code lost or unreadable** | Fallback: type a long phrase. Logged as “phrase confirm”. |
| **Phone restarts overnight** | Android forgets scheduled alarms on reboot. A boot receiver re-reads the saved plans and arms the next ring. |
| **Battery saver kills the app** | The biggest real-world risk. Reliability screen, Home banner, battery exemption request, and on some brands a link to the auto-start page. “Test a real alarm” is the only proof. |
| **You lower the volume** | Volume guard sets the alarm channel back to max while ringing. |
| **Phone dead or off** | Cannot ring. At bedtime, show “Battery 9%, plug in”. |
| **Time zone or clock change** | Listen for time and time-zone changes and recompute the next ring. |
| **App crash mid-ring** | Each ring schedules the next one before it starts, so a crash never breaks the chain. |
| **Emergency stop** | Hold 10 seconds, then type a phrase. Logged, and tomorrow still arms. |

FOR WHEN YOU OPEN THE IDE

## Android build plan

| | |
|---|---|
| **Tools** | **Android Studio** (IntelliJ with Android tooling and the emulator). Kotlin with Jetpack Compose. Your Java skills carry over. These designs use custom tokens, so theme Compose with your own colours, type (Sora and Inter), shapes and spacing rather than stock Material styling. |
| **Ring delivery** | Schedule only the next ring with `AlarmManager.setAlarmClock()`, which is exempt from Doze battery sleep. On fire, a receiver starts a foreground service (media playback type) and posts a full-screen-intent notification that opens the alarm activity over the lock screen. |
| **Sound** | `MediaPlayer` looping on `USAGE_ALARM`. Set the alarm stream to max, then ramp the player volume 0.3 to 1.0 over 60 s with a `ValueAnimator`. Quiet window sets it to 0. |
| **QR scan** | CameraX with ML Kit barcode scanning. The code is a random token you generate, print and stick up. Store its hash. |
| **Motion and haptics** | `animate*AsState`, `AnimatedVisibility`, `Animatable`. Haptics via `HapticFeedbackConstants` (CONFIRM, REJECT, KEYBOARD_TAP). Reduced motion: treat an animator scale of 0 as “off”. Use edge-to-edge and window insets for the system bars. |
| **Storage, permissions** | Room for plans and the morning log. Exact-alarm permission, full-screen intent, notifications, boot completed, wake lock, camera, battery-optimisation request. Sideloading on your own phone avoids Play Store policy limits. |

## Suggested milestones

M1 One hard-coded alarm rings, rising to full volume, over the lock screen on your real phone.



# You’re up.

Awake at 6:12 AM. You stopped 3 rings.

**34 alarms cancelled**
6:15 to 9:00 AM today

**Tomorrow is armed**
Tue · 6:00 AM · Weekdays

Done

# Reliability and sound

## 2 things to fix

Until they're fixed, your alarm may be late or may not ring at all.

### REQUIRED

**Exact alarms**
Rings land on the minute
✓ Allowed

**Full-screen alarm**
Shows over the lock screen
✓ Allowed

**Notifications**
Needed to open the alarm screen
✓ Allowed

**Battery optimisation**
Android may stop WakeUp overnight
Fix

**Auto-start**
Needed on some phone brands
Fix

### OPTIONAL

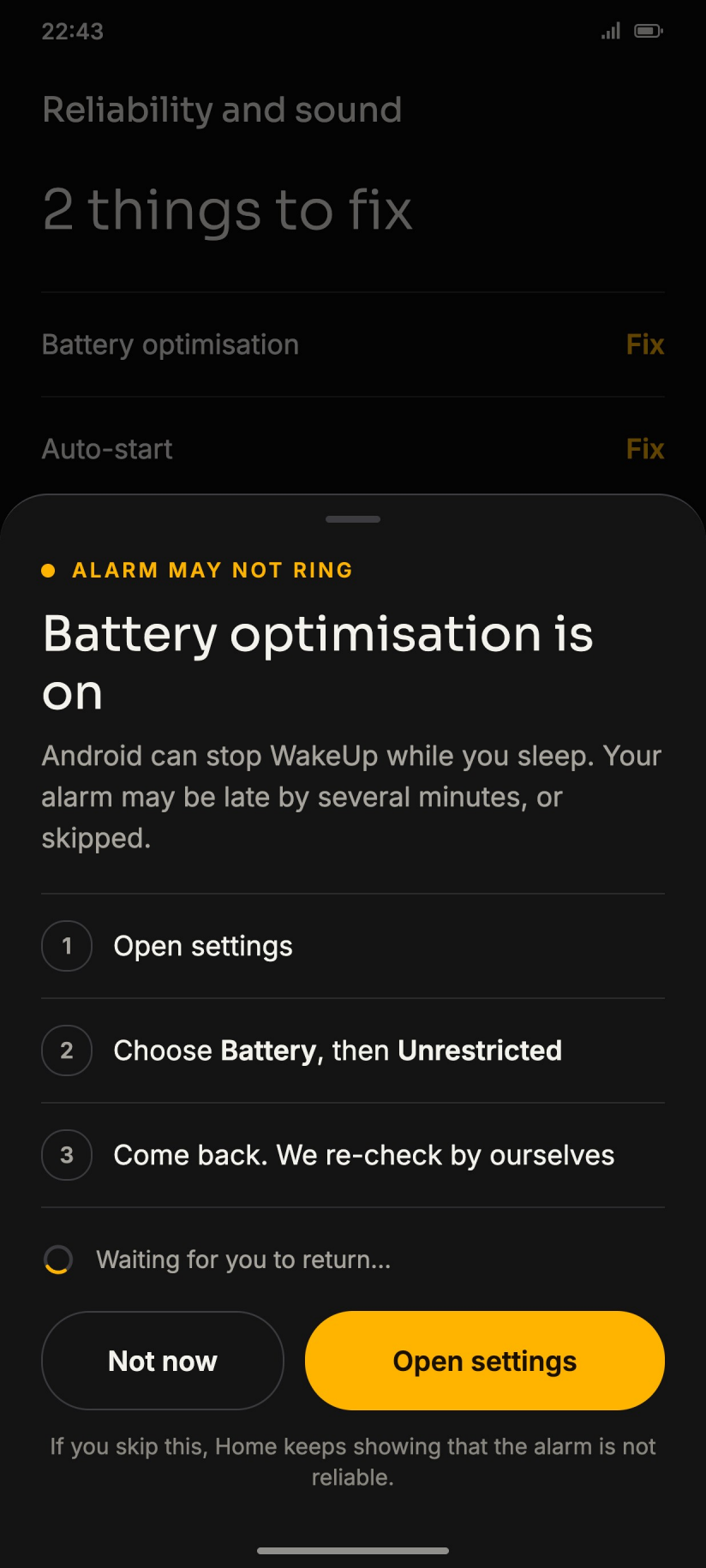
**Alarm sound**
Rising siren
›

**Vibration**
Pattern follows the pulse


**Volume guard**
Restores volume if you turn it down


**Test a real alarm in 1 minute**

Lock your phone first. This is the only proof it works.


22:43
Reliability and sound
2 things to fix
Battery optimisation
Fix
Auto-start
Fix
ALARM MAY NOT RING
Battery optimisation is on
Android can stop WakeUp while you sleep. Your alarm may be late by several minutes, or skipped.
1
Open settings
2
Choose Battery, then Unrestricted
3
Come back. We re-check by ourselves
Waiting for you to return...
Not now
Open settings
If you skip this, Home keeps showing that the alarm is not reliable.

DESIGN SYSTEM · COMPONENTS

# Few parts, used everywhere.

## Buttons

**Primary**
One per screen. Amber, pill, 56 to 72 tall.

Save routine

Scan QR to wake up

Solve to stop

**Secondary, text**
Outline for the other path. Text for low-stakes.

Cancel

Edit routine

Close

**Icon button, FAB**
48dp target. FAB 60, radius 20.

Pressed: Amber pressed. Disabled: 40% opacity.

## List rows and controls

**List rows**
Hairline above, no box. 56 to 72 tall.

First ring 6:00 AM

Vibration
Pattern follows the pulse

Battery optimisation
Android may stop WakeUp overnight
Fix

Exact alarms ✓ Allowed

**Switch**
56 by 48 target, 52 by 32 track. Off, on, on with focus.

**Segmented, day chips**
Native-feeling single choice. Chips for multi-select.

Easy | Medium | Hard

M T W T F S S

Stop a ring | Confirm awake

**Chips**
Status only. Never a button.

Quiet 0:48

Sound on

## Sheets and dialogs

**Bottom sheet**
Radius 28, handle, 24 side padding. Time picker, permission help.

First ring

6 00 AM

Cancel

Set time

Emergency stop?

Hold for 10 seconds, then type the phrase. The morning is logged as an emergency stop.

Keep ringing

Continue

## Alarm states

| STATE | WHAT YOU SEE | SOUND AND ACTIONS |
|---|---|---|
| **Scheduled** | Home hero. A pulsing amber dot only if setup is incomplete. | Silent. Edit, enable or disable. |
| **Ringing** | Full-bleed amber, pulsing rings, 156sp time, two scoped actions. | Rises 30% to 100% over 60 s. Solve to stop, or Scan QR. |
| **Quiet window** | Dark mission screen, chip “Quiet 0:48”, thin countdown line. | Muted up to 60 s. Each correct answer restarts it. |
| **Sound back** | Chip turns amber: “Sound on”. | Resumes at the level it left. Keep solving. |
| **Ring stopped** | Green check, next ring in Display L, amber scan button. | Silent until the next ring. Scan QR or close. |
| **Awake confirmed** | Drawn check, “34 alarms cancelled”, “Tomorrow is armed”. | Silent. Done. |
| **Missed** | Hard stop reached. Home logs “Missed, stopped at 9:00”. | Silent. Tomorrow still arms. |

## Mission states

| STATE | VISUAL | FEEDBACK |
|---|---|---|
| **Idle** | Empty answer box, blinking amber caret. | Keys press down to Surface 2. |